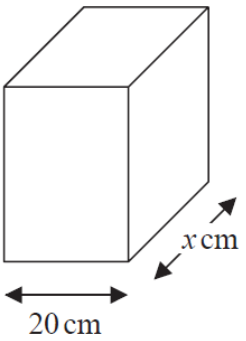
24 The diagram shows a block of metal on horizontal ground.



$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The base of the block of metal is a rectangle 20 cm by x cm.

The block exerts a force of 1500 newtons on the ground.
The pressure on the ground is 3 newtons/cm^2

Work out the value of x.

Area = force ÷ pressure = 1500 ÷ 3

1 mark
Use the pressure formula.

Area = 500 cm^2

20 × x = 500

1 mark
Divide area by length 20.

x = 500 ÷ 20 = 25 cm

1 mark
x = 25.

.....

(Total for Question 24 is 3 marks)